

KISII SCHOOL 2025 PREDICTION



(KCSE PRE-ANALYSIS TEST)

233/3



CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

1. Write your name, admission number in the space provided.
2. Answer all the questions in the spaces provided
3. All working must be clearly shown where necessary.
4. You are not allowed to start working with the apparatus for the first 15 minutes. This time is to enable you read the question paper and make sure you have all the requirements.

FOR EXAMINERS USE ONLY

QUESTION	MARKS	CANDIDATES SCORE
1	24	
2	6	
3	10	
TOTAL	40	

1. You are provided with:

- Magnesium ribbon solid E
- 2M hydrochloric acid, solution F
- 0.375M sodium hydroxide, solution H

You are required to determine

- The rate of reaction between magnesium and hydrochloric acid
- The mass of 1cm length of magnesium ribbon

PROCEDURE 1

Step 1: Cut out five pieces of exactly 1cm length of magnesium ribbon solid E. Carefully fill the burette with solution F.

Step 2: Drain from the burette 10cm³ solution F into a test tube. Put one piece of magnesium ribbon into the tube and immediately start a stopwatch. Record the time taken for the magnesium ribbon to completely react.

Step 3: Transfer all the contents of the test tube into a 250ml volumetric flask. Rinse the test tube with distilled water and put in into the volumetric flask.

Step 4: Repeat step 2 and 3 by placing 9cm³ of solution F and 1cm³ of distilled water shake the contents before use. After each experiment transfer the mixture into the volumetric flask.

Repeat the procedure using contents in test tube 3, 4 and 5 as shown in the table below.

RETAIN THE SOLUTION IN THE VOLUMETRIC FLASK FOR USE IN PROCEDURE II

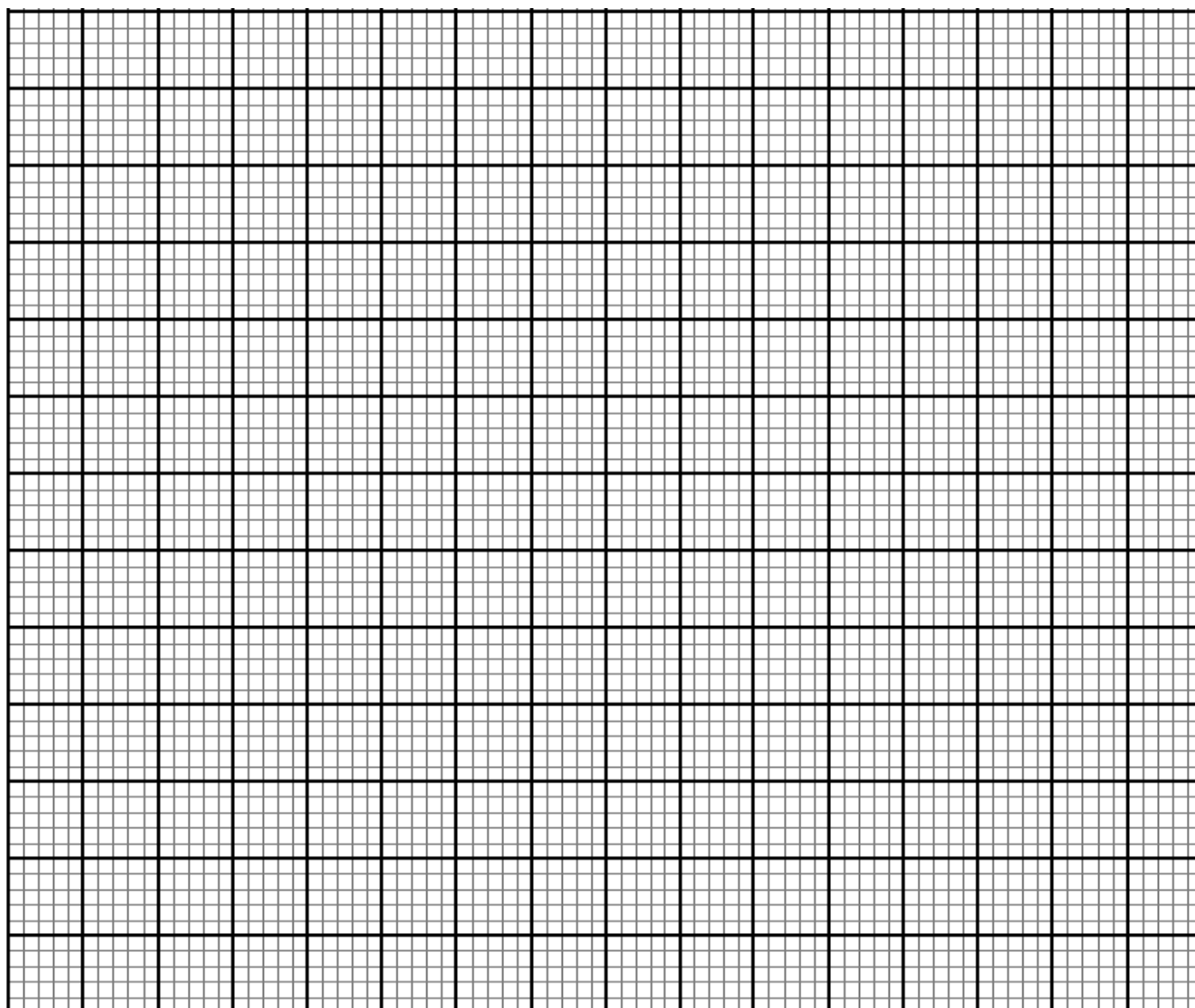
Table I

Test tube number	1	2	3	4	5
Volume of solution F(cm ³)	10	9	8	7	6
Volume of water added (cm ³)	0	1	2	3	4
Time taken (seconds)					
$\frac{1}{t}$ (sec ⁻¹)					

(5mks)

a)i) Plot a graph of rate $\left(\frac{1}{t}\right)$, against volume of solution F.

(3mks)



ii) Use the graph to determine the time taken for 1cm length of magnesium to dissolve if volume of water added is 1.5cm^3 .

(2mks)

iii)a) In terms of rate of reaction, explain the shape of the graph **(2mks)**

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PROCEDURE II

Add distilled water into the solution in the 250ml volumetric flask up to the mark. Label it as solution G.
Clean the burette and fill it with Sodium hydroxide, solution H.

Pipette 25cm³ of solution G into 250ml conical flask. Add 2 drop of phenolphthalein indicator and titrate it with solution H from the burette.

Record your results in table below

Table II

	I	II	III
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of solution H used (cm ³)			

(4mks)

b) Determine the

i) Average volume of solution H used. **(1mk)**

ii) Number of moles of hydrochloric acid in 250cm³ of solution G. **(2mks)**

iii) Number of moles of hydrochloric acid solution F that reacted with solid E. **(2mks)**

- c) To the remaining solution K, add about 10cm³ of dilute hydrochloric acid. Shake thoroughly. Filter the mixture then wash the residue with distilled water. Dry the residue with filter paper.
- i) Place one third of the residue into a test tube. Add about 10cm³ of distilled water and shake the mixture. Add half a spatula sodium hydrogen carbonate.

Observation	Inferences
(½mk)	(½mk)

- ii) To the remaining residue add 5cm³ of distilled water then 5cm³ of sulphuric (VI) acid, followed by 5cm³ of ethanol and warm the mixture.

Observation	Inferences
(1mk)	(1mk)

3. You are provided with solid L carry out the following test and record your observations and inferences in the space provided.
- a) Place one third of the solid L in a dry test tube. Heat it and test any gas produced with blue and red litmus paper.

Observation	Inferences

(2mks)	(1mk)
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b) Place the remaining solid L in a boiling tube. Add about 10cm³ of distilled water and shake. Divide the solution in to four portions.

i) To about 2cm³ of solution L add 2 drops of barium chloride solution

Observation	Inferences
(1mk)	(1mk)

ii) To about 2cm³ of solution L add 5 drops of dilute hydrochloric acid

Observation	Inferences
(1mk)	(½mk)

iii) To 2cm³ of solution L add ammonia solution drop wise until in excess

Observation	Inferences
(1mk)	(1mk)

iv) To 2cm³ of solution L add sodium hydroxide drop wise until in excess

Observation	Inferences

(1mk)	(½mk)
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